

Question 1:

(a)

$$\frac{V_{in}}{RC} = 3.6 \frac{V}{ms} \Rightarrow V_{in} = 3.6 \times 2.5 \times 10^3 \times 2 \times 10^{-6} \times \frac{1}{10^{-3}} = 18 V$$

$$-\frac{V_{ref}}{RC} = -8 \frac{V}{ms} \Rightarrow V_{in} = 8 \times 2.5 \times 10^3 \times 2 \times 10^{-6} \times \frac{1}{10^{-3}} = 40 V$$

(b)

$$t = t_1 + t_2 = t_1 \left(1 + \frac{V_{in}}{V_{ref}}\right)$$

$$\text{Now, } V_1 = 36 = \frac{V_{in}}{RC} t_1 \Rightarrow t_1 = 10 ms$$

$$\therefore t = 14.5 ms$$

$$N = \frac{V_{in}}{V_{ref}} 2^4 - 1 = 6.2. \text{ So, the binary output is } (0110)_2.$$

(c)

$$V_{out} = \frac{R_f}{2R} V_{ref} (0 + 0.5 + 0.25 + 0) = 15 V$$

(d)

$$\Delta = \frac{R_f}{2R} \frac{V_{ref}}{2^3} = 2.5 V$$

$$V_{out_{max}} = \frac{R_f}{2R} V_{ref} (1 + 0.5 + 0.25 + 0.125) = 37.5 V$$

$$(e) \text{ Error} = \frac{V_{in} - V_{out}}{V_{in}} \times 100 = 16.67\%$$

Question 2:

(a)

$$\Delta_{ADC} = \frac{32 - 0}{2^3} = 4 V$$

$$\Delta_{DAC} = \frac{R_f}{R} \frac{V_{ref}}{2^2} = 2 V$$

(b) (i) 4 V — 8 V

(ii) Lowest output from the DAC = 0 V. Step size = 2 V. Thus, outputs within the range in (i) are 4, 6 & 8 V. They are the outputs of '010', '011' & '100' respectively.

(c)

$$R'_f = 2R_f = 40 k\Omega$$

$$\Delta'_{DAC} = \frac{404}{104} = 4 \text{ V}$$

(d) $n_R = 2^4 = 16$, $n_{OA} = 2^4 - 1 = 15$. Encoder: 16 to 4.

Question 3:

(a) $V_{TH} = \frac{3}{2+3}10 = 6 \text{ V}$, $V_{TL} = \frac{3}{2+3}(-10) = -6 \text{ V}$

$$T_1 = R_x C_x \ln \left(\frac{-6 - 10}{6 - 10} \right) = R_x C_x \ln(4) \Rightarrow \frac{1}{2 \times 1000} = 10 \times 10^3 \ln\left(\frac{7}{3}\right) \times C_x \Rightarrow C_x = 36 \text{ nF}$$

$$\tau = R_x C_x = 0.36 \text{ ms}$$

(b)

$$T_1 = R_x C_x \ln \left(\frac{V_{TL} - V_H}{V_{TH} - V_H} \right) = R_x C_x \ln \left(\frac{\frac{R_1}{R_1 + R_2} V_L - V_H}{\frac{R_1}{R_1 + R_2} V_H - V_H} \right)$$

$$= R_x C_x \ln \left(\frac{\frac{R_1}{R_1 + R_2}(-V_H) - V_H}{\left(\frac{R_1}{R_1 + R_2} - 1 \right) V_H} \right) = R_x C_x \ln \left(\frac{-\left(\frac{R_1}{R_1 + R_2} + 1 \right)}{\frac{R_1}{R_1 + R_2} - 1} \right) = T_2$$

$$T = 2T_1 = 2R_x C_x \ln \left(\frac{2R_1 + R_2}{R_2} \right)$$

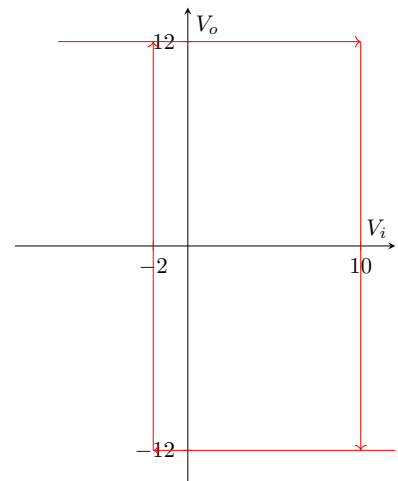
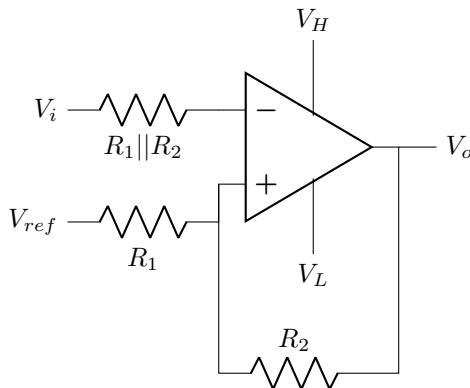
$$\Rightarrow T_{new} = 2R_x C_x \ln \left(2\frac{R_1}{R_2} + 1 \right)$$

$$\Rightarrow \frac{1}{f_{new}} = \frac{1}{2f} = \frac{1}{2 \times 1000} = 2 \times 0.36 \times 10^{-3} \ln \left(2\frac{R_1}{R_2} + 1 \right)$$

$$\Rightarrow \frac{R_1}{R_2} = 0.5$$

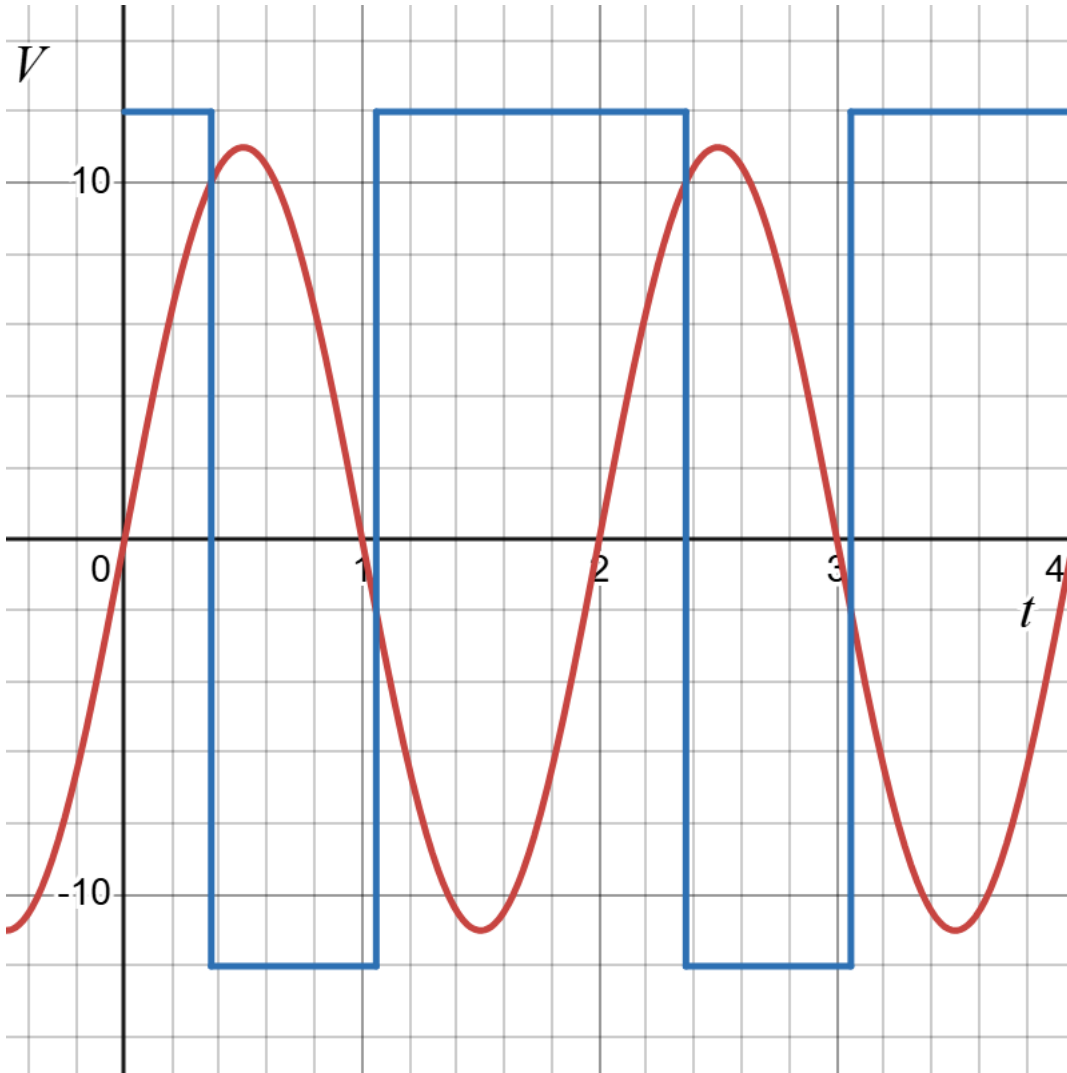
(c) Inverting. $V_H = 12 \text{ V}$, $V_L = -12 \text{ V}$, $V_{TH} = 6 \text{ V}$, $V_{TL} = -6 \text{ V}$, $V_S = 0 \text{ V}$, $V_{HW} = 12 \text{ V}$

(d)



$$R_1 = 1\text{ k}\Omega, R_2 = 1\text{ k}\Omega, V_{ref} = \frac{4}{\frac{1}{1+1}} = 8\text{ V}, V_H = 10\text{ V}, V_L = -10\text{ V}$$

(e)



Question 4:

(a) $T = 6\text{ ms}$

$$V_{TH} = 3.5 = -\frac{R_1}{R_2}(-7) \Rightarrow \frac{R_1}{R_2} = \frac{1}{2}$$

$$V_{TL} = -\frac{1}{2}8 = -4\text{ V}$$

$$\therefore D = \frac{T_2}{T_1 + T_2} = \frac{\frac{V_{TL} - V_{TH}}{V_L}}{\frac{V_{TH} - V_{TL}}{V_H} + \frac{V_{TL} - V_{TH}}{V_L}} = 53.33\%$$

$$T_2 = D \times 6 \times 10^{-3} = 3.2\text{ ms}$$

$$C_i = \frac{T_2}{R_i \frac{V_{TL} - V_{TH}}{V_L}} = 1.49\text{ }\mu\text{F}$$

(b) $D_{sqr} = (1 - D_{tri}) \times 100 = 46.67\%$

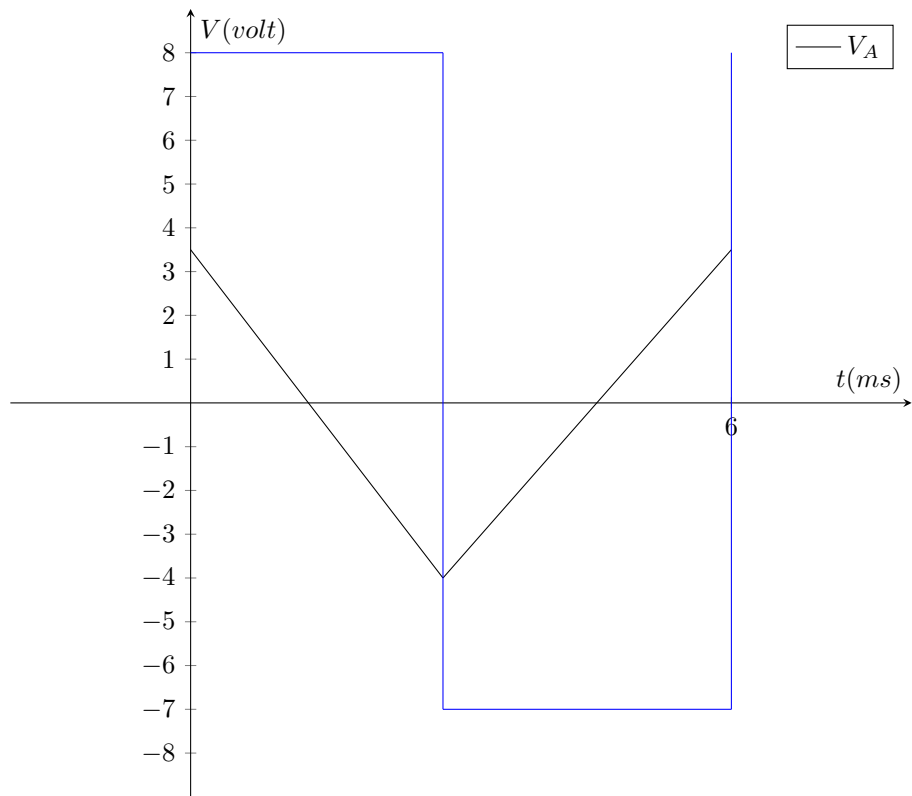
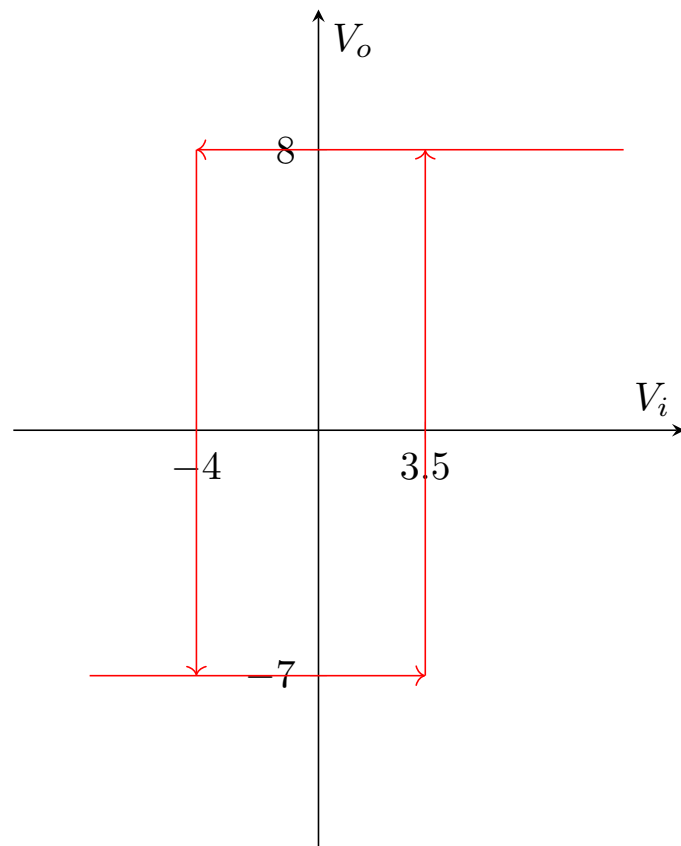
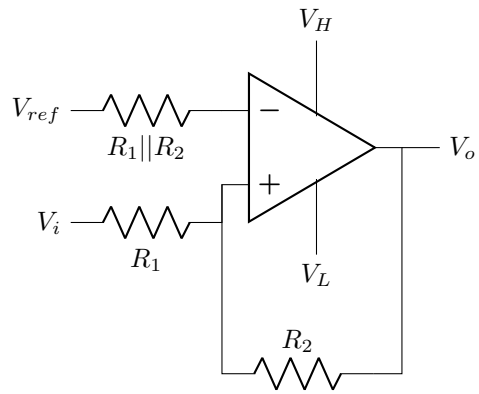


Figure 1: Output of the Triangular Wave Generator

(c)



(d)



$$V_H = 8\text{ V}, V_L = -7\text{ V}, R_1 = 1\text{ k}\Omega, R_2 = 2\text{ k}\Omega, V_{ref} = \frac{-2}{\frac{1+2}{2}} = -1.33\text{ V}$$

